HOMEWORK E.3 · EOG REVIEW · NC.8.EE.7 · NC.8.EE.4 · NC.8.G.5 · NC.8.EE.8

EOG Review: Equations & inequalities

May 17 / May 18 · Targets: EE.7-1, EE.7-2, EE.7-3, EE.7-4, EE.4-2, EE.7-7, EE.7-6, G.5-6, EE.7-9, EE.8-9

Name _____ Date _____ Block _____

Remember: Undo the operations in reverse order. Distribute first, combine like terms, then get the variable on one side. If you multiply or divide by a **negative**, **flip** the inequality sign: $-3x > 12 \rightarrow x < -4$. **Solutions**: $x = 5$ → one; 4 = 4 → infinitely many; 4 = 7 → none.

SCORE 3.0 Approaching · the basics

Targets: EE.7-1, EE.7-2, EE.7-3, EE.7-4, EE.4-2

1 Solve: $\frac{x}{4} + 6 = 1$ EE.7-1

2 Solve: $2(3x - 1) + 4x = 28$ EE.7-2

3 Solve: $9x + 5 = 4x - 15$ EE.7-3

Ⓐ $x = -4$ Ⓑ $x = 4$ Ⓒ $x = -2$ Ⓓ $x = 2$

4 How many solutions? EE.7-4

$4(2x - 1) + 1 = 8x - 3$

One None Infinitely many

5 Divide. Write the answer in scientific notation: $(8.4 \times 10^6) \div (2.1 \times 10^2)$ REVIEW 2.7 EE.4-2

SCORE 4.0 Proficient · on target

Targets: EE.7-7, EE.7-6, G.5-6

6 Solve and graph: $5 - 3x < 17$ EE.7-7

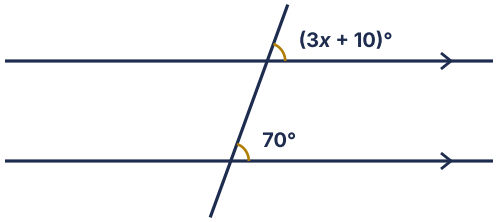


7 Which equations have **no solution**? EE.7-6

- ☐ $2x + 5 = 2x - 5$
- ☐ $3(x - 1) = 3x - 3$
- ☐ $4x + 1 = x + 1$
- ☐ $6 - x = -x + 7$
- ☐ $2(x + 4) = 2x + 8$

Select all that apply.

8 The lines are parallel. Find x. REVIEW 4.3 G.5-6



SCORE 5.0 Mastery · go deeper

Targets: EE.7-9, EE.8-9

9 Jamal has $120 and saves $15 each week. Priya has $60 and saves $25 each week. When will they have the same amount? What does your answer mean? EE.7-9

My equation is _____ . After _____ weeks, they each have $_____ .

10 A system is $y = 2x + 1$ and $y = 2x - 3$. How many solutions does it have? Explain using the graphs. REVIEW 6.12 EE.8-9

The slopes are _____ and the y-intercepts are _____ . So the lines _____ .